# Diversify the Decisions, Not Just the Assets

## A Practical Architecture for Dynamic Asset Allocation When Strategy Choice Is Uncertain

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### Abstract

Diversification usually starts with assets. This paper asks whether it should also reach the decisions that choose them. Dynamic asset allocation can respond to changing market evidence without requiring a precise forecast of the next regime, but many practical rules descend from the same momentum intuition. A portfolio can therefore hold several strategies and still depend on one broad decision.

We examine diversification at the decision level: What does each rule own? When does it change? How much risk does it take? This project uses the name Autonomous Dynamic Asset Allocation (ADAA) for the rule-governed architecture studied here. ADAA is treated as a worked implementation, not a timeless recipe: its five current sleeves are practitioner adaptations of public rule families inside a modular portfolio architecture. The analysis separates source-family selection from sleeve engineering, then examines transition timing, persistence, top-level weight uncertainty, trading costs, drawdowns, stress periods, and explicit failure modes.

In a June 2008-July 2026 historical simulation on a common ETF sample, the current sleeves make meaningfully different portfolio decisions even when some return correlations are high. A separate performance-free decision check reconstructs a 16-rule 2023 reference pool: the documented historical five-family set ranks above most feasible five-rule combinations in that pool, while the same check also exposes redundancy inside the original design. The later practitioner weights sit on a broad high-performing plateau rather than at a unique backtest optimum. ADAA does not beat the hindsight-best sleeve and does not protect in every stress episode. The evidence supports a narrower proposition: diversify reasonable decision rules, avoid precision unsupported by the data, make failure modes visible, and treat sleeves as replaceable components.

## 1. Start With a Response, Not a Forecast

Dynamic allocation starts from a simple ambition: take less risk when the evidence weakens without pretending to know exactly when the next regime begins.

Momentum is one practical answer. It replaces the question, 'What will happen next?' with a more observable one: 'Is this exposure still being rewarded, or has the evidence changed?'

> The aim is not to call the turn. It is to decide in advance how the portfolio will respond when the evidence changes.

A systematic rule cannot see the future either. Its advantage is narrower and more useful: the response is specified before the next market story arrives. That makes the process easier to review, limits improvisation after the fact, and makes failure easier to diagnose.

In ADAA, 'autonomous' means rule-governed rather than discretionary: the response rules are specified in advance. It does not imply a self-learning or self-optimizing system, and it does not remove the need for human governance.

There are several plausible reasons trends can persist. Information reaches investors at different speeds. Institutions adjust large portfolios gradually. Economic and policy regimes can last for months or years rather than days. Investors may underreact at first and extrapolate later. The argument in this paper does not depend on any one explanation.

Foundational research documents momentum in equities and across asset classes (Jegadeesh and Titman 1993; Moskowitz, Ooi, and Pedersen 2012; Asness, Moskowitz, and Pedersen 2013), while behavioral and information-diffusion models offer possible explanations for its persistence (Barberis, Shleifer, and Vishny 1998; Hong and Stein 1999).

That intuition has produced a large family of practical allocation rules. Faber uses simple trend rules to alter risky-asset exposure. Antonacci combines relative and absolute momentum. Keller and collaborators add canary assets, defensive universes, breadth, volatility, and correlation inputs. Adaptive Asset Allocation takes a broader portfolio-construction view and changes portfolio inputs as market conditions change (Faber 2007; Antonacci 2017; Butler et al. 2012; Keller and van Putten 2012; Keller 2022; Keller and Keuning 2023).

Despite the different mechanics, the common discipline is to let observed market behavior change risk rather than hold the same allocation through every environment.

Every adaptive rule also has to choose a speed. Too slow and it can give back protection. Too fast and it can get whipsawed. Defensive at the wrong moment, it can miss the first stage of a sharp rebound.

Momentum's success creates a second uncertainty: model choice. Once a useful idea is widely understood, it can be expressed through many reasonable lookbacks, universes, canary rules, defensive assets, breadth filters, and weighting schemes. The menu grows, but certainty does not. Part of the problem moves from forecasting the market to choosing among models: which reasonable expression of momentum will work best in the next regime?

That model-selection concern also appears in the author's private June 2023 university project report documenting an earlier ADAA design. The report is used only as a record from that time for the problem formulation and historical strategy set. Section 4 separates that history - including the later adoption of HAA - from the present empirical validation.

Momentum can therefore be useful and still create concentration: several rules built on the same successful idea may end up making much the same decision. More strategy names do not necessarily create more decision diversity. The practical question is whether the rules actually disagree when it matters.

## 2. Five Strategies Can Still Be One Decision

Imagine a portfolio with five dynamic-allocation strategies. On paper it looks diversified: five models, five rule sets, five names.

Now imagine that all five depend heavily on momentum, turn defensive in roughly the same month, and return to risk after roughly the same signal. The investor owns five strategies but may still be making one underlying decision.

The analogy is familiar. Owning five technology stocks is not the same as owning five independent sources of risk. Strategy diversification can fail in the same way.

Return correlation is a useful first check, but it describes outcomes more directly than decisions. Two rules can earn together while changing portfolios at very different times, or hold different assets while de-risking on the same date.

This framing builds on substantial precedent. Forecast-combination and thick-modeling research studies diversity across forecasts or models (Batchelor and Dua 1995; Granger and Jeon 2004; Aiolfi and Favero

2005; Pesaran, Schleicher, and Zaffaroni 2009). Portfolio research warns against overconfidence in precisely estimated optimal weights (Michaud 1989; DeMiguel, Garlappi, and Uppal 2009), while practitioner work compares strategy correlations and overlapping positions (Ludlow 2022). The paper's narrower contribution is a transparent way to examine decision-level diversification in dynamic asset-allocation rules.

We describe each rule in three simple dimensions:

- **What?** Which assets does the rule choose?
- **When?** When does it change its mind?
- **How much?** How much risk or defensive exposure does it take?

These three questions turn an abstract idea - strategy diversification - into observable behavior. In the formal Decision-Space check, target-weight distance captures What and How Much together, holdings overlap adds a separate composition check, and transition-date overlap captures When. The strategy names can change; the questions still work.

A slow rule can diversify a fast one even when their returns are moderately correlated. Two rules can own different assets yet de-risk on the same dates. A portfolio can therefore be diversified in assets and concentrated in its decision clock.

ADAA earns decision diversification only if the sleeves contribute meaningfully different decisions - and only if the architecture can preserve that disagreement as its components evolve.

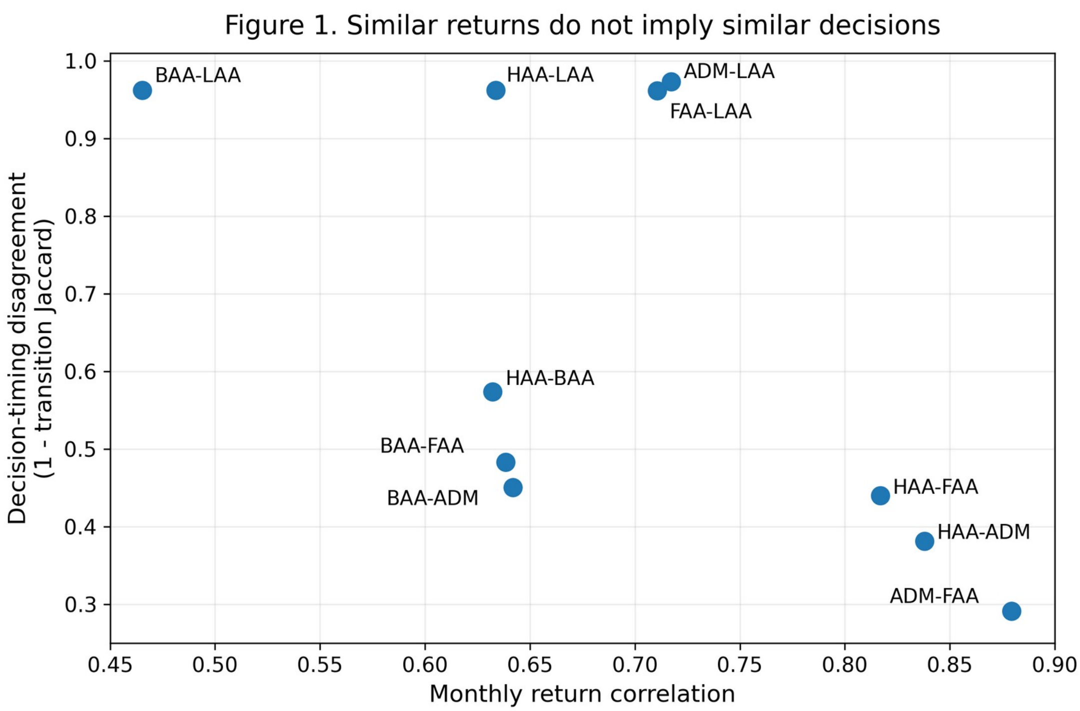


**Figure 1. Similar returns do not imply similar decisions.** The horizontal axis shows monthly return correlation. The vertical axis shows disagreement in portfolio-transition timing, measured as one minus the share of transition months common to both strategies (Jaccard overlap). A value near one means the pair rarely changes its target weights in the same month. Return correlation remains useful, but it does not fully describe how rules behave.

## 3. A Strategy Is More Than Its Tickers

A published rule is a blueprint, not a production portfolio. New ETFs appear; some original instruments are unavailable, expensive, or narrower than the intended exposure. The practical difficulty is that changing the investment expression can also change the strategy itself.

That distinction matters because the current ADAA sleeves are not presented as untouched copies of published strategies. They are practitioner adaptations inspired by public rule families.

The audit therefore asks three questions: What did the source rule actually do? What changed in the ADAA implementation? And did the adaptation preserve the role that made the source rule useful in the first place?

The trap is familiar: modify a rule or universe, observe a better history, and then use that same history as the justification for the modification.

The standard here is stricter: an implementation change should be described as role-preserving only when the underlying decision role survives; otherwise it should be called a redesign.

Table 1 summarizes the current classification. The public replication package retains the fuller source history, the reasons for the modifications, and the measured parent-versus-variant effects; the main-text version is deliberately compact.

***Table 1. From published rules to investable sleeves***

| Sleeve | Public source family | Main ADAA adaptation | Classification | What the audit says |
|---|---|---|---|---|
| HAA | Keller & Keuning HAA | Wider offensive universe, Top 6, modified replacement logic | **Material practitioner variant** | The TIP-canary regime clock is preserved, but holdings breadth and portfolio expression change materially. |
| BAA Aggressive | Keller BAA Aggressive | Mainly VEA/VWO/BND → EFA/EEM/AGG expression changes | **Near-parent expression variant** | The current sleeve exactly matches the parent rule on the ADAA expression; the source and ADAA expressions produce different states in 12 of 216 months. |
| ADM | Hanly Accelerating Dual Momentum | Narrow one-position selector → broad multi-asset Top-6 sleeve | **Major architecture variant** | The role itself is broadened; this should not be presented as source-faithful ADM. |
| FAA | Keller & van Putten FAA | Larger universe, 3/6/12 momentum structure, peer-only correlation, exact Top 6 | **Major rule + universe variant** | Both rule and universe changes materially alter decisions. |
| LAA | Keller LAA | Persistent core retained; IWD expression → SPY/EEM/EWY mix; timing information state clarified | **Role-preserving persistent variant** | What changes, while the slow transition clock is largely preserved. |

Source note: full source history, reasons for the modifications, and parent-versus-variant decision effects are supplied in the public replication package; the main-text table is intentionally compact.

A strategy is not its ticker list, but the ticker list is not irrelevant either. We therefore separate the choice of a public strategy family from the engineering that turns it into an investable ADAA sleeve. Where the comparison is meaningful, we first hold the parent rule fixed and change only the universe; then we hold the universe fixed and change the rule. Where that two-step comparison is not meaningful - as with the narrow parent ADM versus a broad Top-6 multi-asset rule - we do not force a mechanical 2x2 comparison.

***Table 1B. Universe and rule engineering can change the decision***

| Sleeve | Cleanest engineering comparison | Structural result | Interpretation |
|---|---|---|---|
| HAA | Parent rule: parent vs ADAA universe | L1 0.563; timing agreement 0.692 | The wider universe changes the parent rule materially before Top-N/replacement changes are applied. |
| BAA Aggressive | Parent rule: source vs ADAA ETF expression | L1 0.275; timing agreement 0.814 | Mostly role-preserving, but ETF substitution is not decision-neutral. |
| ADM | Parent architecture vs ADAA architecture | L1 2.000; timing agreement 0.314 | Major redesign; a forced mechanical 2×2 would be misleading. |
| FAA | Parent rule: parent vs ADAA universe | L1 1.336; timing agreement 0.686 | Universe expansion alone materially rewires decisions; later rule changes add another material layer. |
| LAA | Parent rule: parent vs ADAA equity expression | L1 0.500; timing agreement 1.000 | Holdings change while the parent transition clock is preserved exactly. |

Structural note: L1 is the average sum of absolute target-weight differences after mapping equivalent ETF exposures to common economic exposures (range 0-2). Appendix A divides this quantity by two for its 0-1 Decision-Space scale. Timing agreement is the Jaccard overlap of transition dates. No return statistic enters the table.

The LAA comparison is especially clean. Replacing the original equity expression changes what the sleeve owns, but under the parent timing rule the QQQ/SHY switching state is unchanged across the common sample. The holdings expression changes; the decision clock does not.

ADM is the opposite case. The current ADAA-ADM differs too much from the narrow parent strategy in its portfolio rules to be defended as a simple implementation update. It is more accurately described as a practitioner redesign inspired by the source idea.

The purpose of the audit is to identify each modification clearly and trace what it changed, rather than force every adaptation into a source-faithful label.

## 4. Different Rules Need Different Jobs

The current ADAA implementation combines five sleeves: HAA, BAA Aggressive, ADM, FAA, and LAA. In plain English, their jobs differ:

- **HAA —** A canary-gated momentum sleeve: defensive when its regime signal weakens, broader offensive selection when risk is on.
- **BAA Aggressive —** An explicit offensive/defensive regime switch between risk-on candidates and a defensive universe.
- **ADM —** A fast relative-momentum Top-6 selector across a broad multi-asset universe.
- **FAA —** A broad cross-sectional sleeve combining momentum with volatility and correlation information.

- **LAA —** A deliberately persistent sleeve with a much slower switching clock.

The five sleeves were not discovered by an algorithm. They reflect practitioner judgment applied to a public strategy landscape. The private June 2023 report documents three parts of that intent: reliance on rules rather than subjective market views; concern about common momentum dependence; and use of a slow-moving sleeve with explicit drawdown awareness. Because the report was private, it is a record from that time, not preregistration or forward validation. Some ticker substitutions and later replacements are documented less completely and remain author recollection.

The original judgment was that spreading reliance across several reasonable rules was more defensible than concentrating in whichever backtest looked best. That explains the architecture; it does not validate it. The current sleeves are one implementation generation, and they can be challenged or replaced through explicit tests of role, redundancy, robustness, and implementation.

The sleeves can change. The architecture is the part meant to survive.

A fair criticism is that ADAA may be only a hand-built bundle of familiar rules. The paper tests that concern in three stages:

1. Reconstruct source-anchored public strategy families and compare them in a Strategy Zoo that does not use performance data.
2. Audit separately how each family was engineered into the investable sleeve.
3. Evaluate the actual ADAA variants together as a portfolio.

This ordering avoids a circular argument in which a modified universe is made different and then used as proof that the original family selection was diverse.

The separation reveals an important trade-off. Using the same Decision-Space distance measure, which does not use returns or other performance statistics, the five source-family representations corresponding to the 2023 historical set have an average pairwise distance of about 0.82. Their current ADAA practitioner expressions are about 0.64, roughly 22% lower.

The same pattern appears in the current set: about 0.85 for the source-family parents versus 0.66 for the actual current variants. Much of the compression comes from ADM and FAA, whose broad multi-asset redesigns make them substantially more alike.

That compression is not automatically a design failure. A sleeve can become more diversified internally, less concentrated, or easier to implement while becoming more similar to the other sleeves. The portfolio problem therefore has two levels: diversify within each strategy, while preserving useful disagreement between strategies.

The measure is meant to make hidden concentrations visible enough to govern as the components change; it is not an automated strategy-selection rule.

Figure 2 summarizes the five current sleeves as decision fingerprints. The horizontal axis captures what by showing the average number of active holdings. The vertical axis captures when by showing how often target weights change. Marker size represents the average rule-defined defensive allocation, a simple view of how much risk the rule removes.

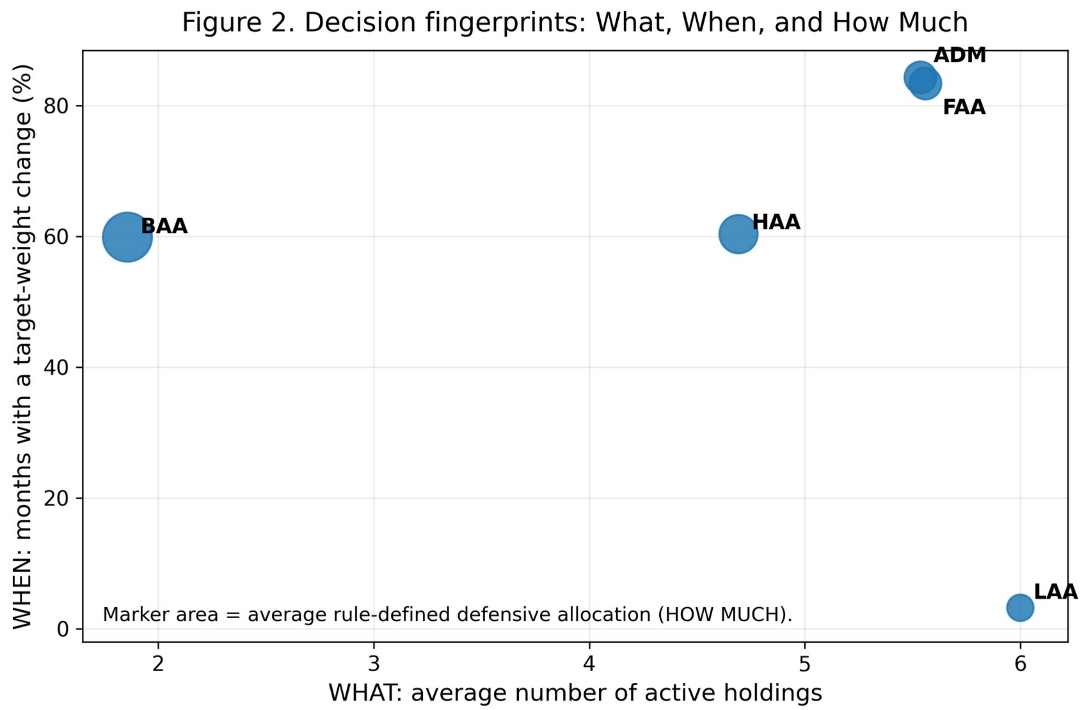


**Figure 2. Decision fingerprints: What, When, and How Much.** The dimensions are descriptive rather than a single optimized score. Their purpose is to reveal which kinds of disagreement each sleeve adds.

Figure 2 makes LAA's distinct clock easy to see. It does not make LAA a crash hedge: in the observed sample it did not always reduce drawdowns and was a material drag in 2022. Its contribution is better described as persistence - a different decision clock - than as guaranteed protection.

## 5. The Current ADAA Implementation in One Page

The mechanics fit on one page, which matters because a process should be understandable enough to challenge. These rules describe the current generation of ADAA, not an immutable specification.

1. Each sleeve observes its own signals at month-end.
2. Each sleeve produces a transparent target portfolio.
3. The five target portfolios are combined at the underlying-asset level.
4. New month-end targets become effective in the next holding period.
5. Trading costs are applied after offsetting sleeve-level trades at the final account level. A buy in one sleeve and a sell in another are therefore not charged separately when they cancel each other.

The mechanism study uses equal 20% sleeve weights so the main result does not depend on a fitted top-level allocation. Separately, we report the later practitioner weights used by the author: HAA 25%, BAA Aggressive 15%, ADM 17.5%, FAA 17.5%, and LAA 25%. These are not the June 2023 weights, and no written record from that time documents the exact rule used to choose them. Section 7 therefore treats them as a practical comparison case, not as a research design set in advance or an optimal allocation.

A robust architecture should not require too many uncertain choices to be exactly right at once. If success depends on identifying both the future winning rule and its exact weight, the design is asking too much of foresight.

The next sections therefore separate two uncertainties: which reasonable rule will work best, and how precisely the rules need to be weighted.

## 6. Do the Rules Actually Disagree?

They do, but unevenly. That is why disagreement has to be measured rather than assumed.

Some pairs are clearly close. ADM and the current FAA sleeve have a gross monthly return correlation of about 0.88, the highest among the five current sleeves, and their average target-weight distance is also relatively small. They are the clearest redundancy candidate in the present architecture. That diagnosis is not used to replace either sleeve after seeing the results; any future replacement test should be defined before its results are known and evaluated separately.

Other pairs show why return correlation is not enough. HAA and LAA have a monthly return correlation of about 0.63, but their transition-timing disagreement is about 0.96. ADM and LAA have a return correlation above 0.71, yet their timing disagreement is about 0.97. The rules can earn together while changing at very different times.

Returns describe realized outcomes, while transition overlap reveals part of the process that generated them. The two measures answer different questions, and Figure 3 makes the timing dimension concrete.

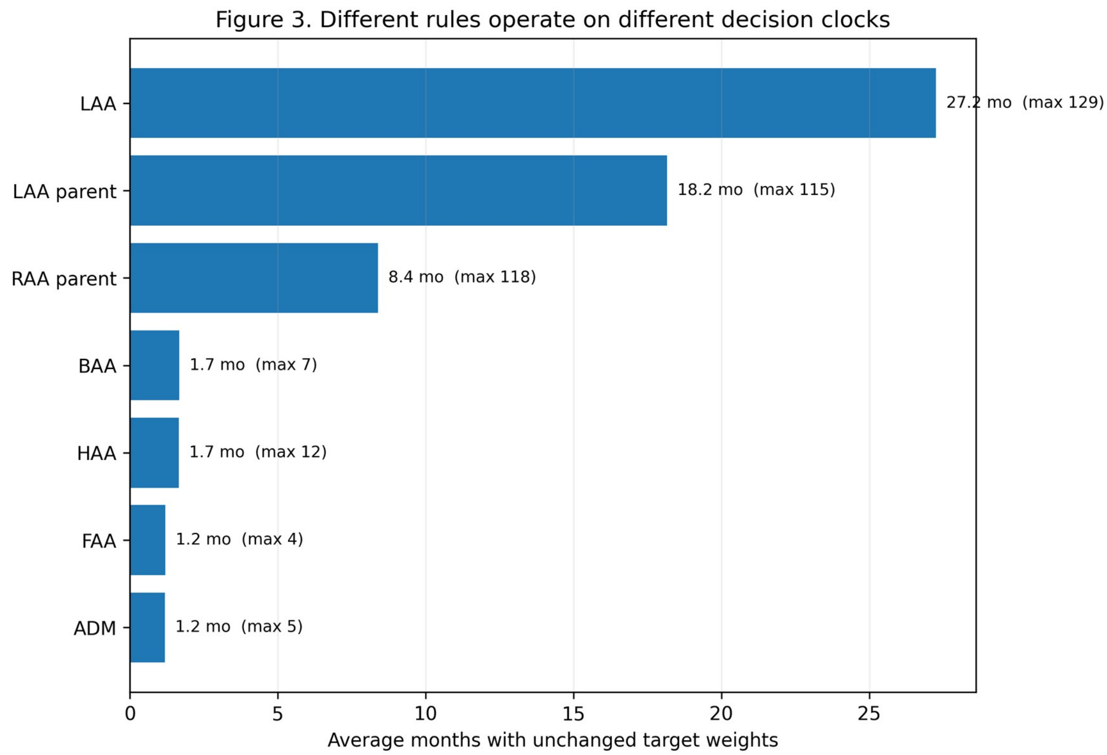


**Figure 3. Different rules operate on different decision clocks.** The current HAA and BAA targets remain unchanged for roughly 1.7 months on average; ADM and FAA for about 1.2 months. Current ADAA-LAA remains unchanged for about 27 months on average. The source LAA and the out-of-portfolio RAA comparator are also materially more persistent than the faster dynamic sleeves.

RAA is useful as an additional check. It is not an independent source family - it also comes from the Keller strategy family - but it shows that slow decision making is not unique to the current ADAA-LAA implementation. Similar persistence in source LAA and RAA supports the broader idea that a slow-moving sleeve can add a distinct decision-speed dimension.

That does not mean every ensemble needs LAA, or that persistence is always beneficial. Persistence is a portfolio property, not a permanent job assignment. A future architecture can seek the same property through a different sleeve if the evidence warrants it.

Even after the decision roles are visible, another uncertainty remains: which reasonable rule will look best next? Figure 4 shows why hindsight is a difficult standard. The identity of the best standalone sleeve changes as the trailing window moves. HAA is the most frequent 60-month Sharpe winner, BAA also leads for long stretches, and LAA briefly leads in earlier windows. ADM and FAA never lead on this particular 60-month metric.

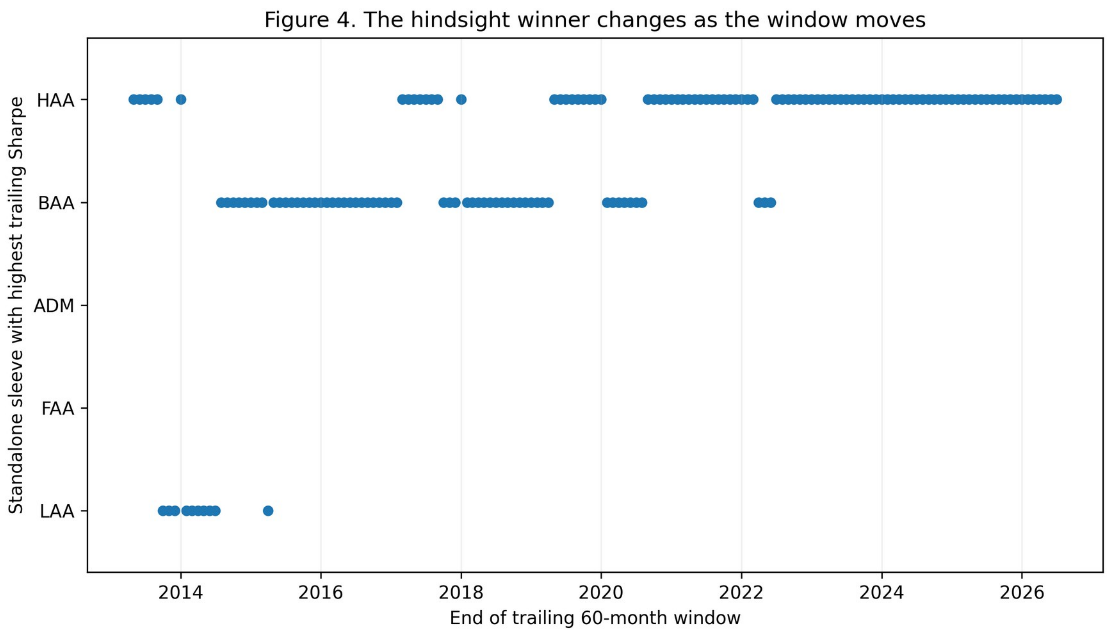


**Figure 4. The hindsight winner changes as the window moves. Each point identifies the standalone sleeve with the highest trailing 60-month BIL-excess Sharpe. The chart is not a timing strategy and does not allow ex-post switching. It illustrates model uncertainty: the rule that looks best depends on where the evaluation window ends. ADAA using the later practitioner weights is above the median constituent in 98.7% of 60-month Sharpe windows, top-two in 69.2%, and bottom-two in 0%.**

> A multi-rule portfolio need not beat the best rule after the fact. Its practical role is to reduce the cost of not knowing beforehand which reasonable rule will be best, while remaining governable when that answer changes.

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## 7. Do Not Chase the Peak; Look for the Plateau

The June 2023 project did use optimization. It explored 100,000 Monte Carlo weight vectors and reported maximum-Sharpe and minimum-variance allocations subject to a 10% minimum sleeve weight. The later practitioner weights used in this paper are different, so the historical project should not be described as optimization-free.

There is no written record from that time explaining the exact later weighting rule. The author recalls choosing a moderate compromise between equal weighting and past-performance considerations rather than carrying forward a precise optimizer solution. Because that explanation is a later recollection, the weight choice itself is not treated as research evidence.

On the current sample, an ex-post maximum-Sharpe search with a minimum 10% allocation to each sleeve would put roughly half the portfolio in HAA and the minimum 10% in several other sleeves. That is not the later practitioner allocation. Yet the later practitioner weights capture about 96.6% of the ex-post maximum gross Sharpe, and equal 20% weights capture about 95.2%. Across 100,000 weight vectors that satisfy the minimum-weight constraint, about 40% achieve at least 95% of the ex-post maximum Sharpe.

The observed landscape is broad rather than narrowly peaked: many materially different weight vectors sit close to the ex-post peak.

The peak also moves. In rolling 60-month optimizations, HAA's optimal weight ranges from the 10% floor to the 60% cap; BAA does the same; LAA ranges from the floor to above 50%. Block-bootstrap resampling produces similarly wide ranges.

Figure 5 combines the two ideas. The circles-and-whiskers show sleeve-weight ranges among the 100,000 feasible portfolios that retain at least 95% of the full-sample maximum Sharpe. The square-and-whisker ranges show where the ex-post optimum moves under block resampling. The later practitioner weights are neither the full-sample optimum nor a special narrow optimum.

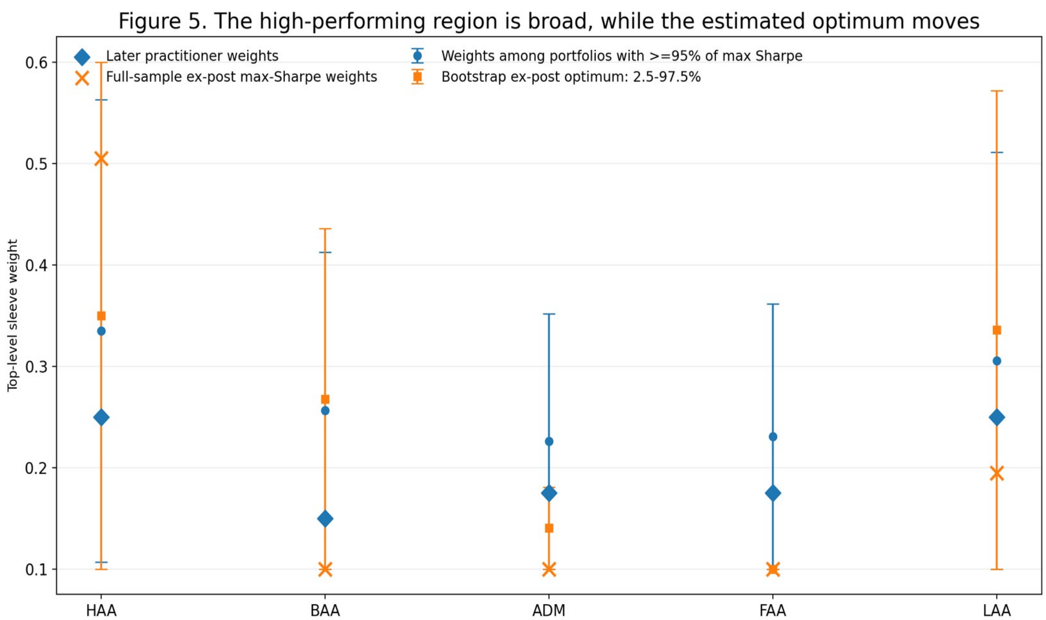


Figure 5. The high-performing region is broad, while the estimated optimum moves. The near-optimal ranges are descriptive properties of the fixed full sample, not confidence intervals. The bootstrap ranges show how much the ex-post optimum moves when the historical path is resampled in 12-month blocks. About 40.3% of the 100,000 weight vectors satisfying the minimum-weight constraint retain at least 95% of the full-sample maximum Sharpe.

The lesson is not to reject optimization. It is to reject the confidence implied by a precise historical optimum when many nearby allocations perform similarly and the estimated peak moves with the sample.

> Avoid the cliffs; do not chase the peak.

That is a design principle, not a statistical guarantee. The evidence here says only that observed ADAA performance is relatively insensitive across a broad neighborhood of top-level weights while the estimated optimum moves substantially.

A further robustness check, designed after the main performance results had been examined, reaches the same practical conclusion. A rolling 60-month maximum-Sharpe control uses the prior 60 completed holding-period returns, keeps each sleeve between 10% and 60%, and applies the estimated weights in the next month. It earns slightly more gross CAGR but trades more; after a 25-basis-point one-way cost assumption, the fixed practitioner allocation has the higher CAGR and Sharpe. Because this control was specified after the main performance results were known, it is an adversarial diagnostic, not confirmatory evidence or a candidate replacement. It does not prove that fixed weights always beat rolling optimization; it shows that chasing a moving optimum is not free.

The portfolio philosophy therefore has two layers:

- **Decision Diversification** addresses uncertainty about which rule will work best.
- **Weight Robustness** addresses uncertainty about how precisely those rules should be combined.

## 8. What Does Decision Diversification Look Like in the Data?

The primary investable sample runs from June 2008 through July 2026, using only ETFs available for the five-sleeve architecture and the required lookback histories. The start date is data-defined rather than selected from portfolio performance. Because several source rules and current variants postdate June 2008, the evidence is a historical common-sample simulation, not a live or post-publication track record.

The ADAA portfolio using the later practitioner weights produces the following results over the 218-month sample:

***Table 2A. Core observed performance and implementation costs***

| Portfolio | Basis | CAGR | Volatility | BIL-excess Sharpe | Maximum drawdown |
|---|---|---|---|---|---|
| ADAA — practitioner weights | Gross | **10.80%** | **8.97%** | **1.05** | **-10.34%** |
| ADAA — practitioner weights | 25 bps one-way | **9.36%** | **8.97%** | **0.90** | **-10.75%** |
| ADAA — equal 20% | Gross | 10.66% | 8.97% | 1.03 | -10.00% |
| ADAA — equal 20% | 25 bps one-way | 9.07% | 8.97% | 0.87 | -10.43% |
| HAA — hindsight strongest sleeve | Gross | 12.97% | 10.70% | 1.07 | -11.68% |
| HAA — hindsight strongest sleeve | 25 bps one-way | 11.20% | 10.76% | 0.92 | -12.30% |
| 60/40 SPY/IEF | Gross | 8.44% | 9.72% | 0.75 | -27.55% |
| 60/40 SPY/IEF | 25 bps one-way | 8.38% | 9.71% | 0.75 | -27.61% |
| SPY | Gross | 11.68% | 15.73% | 0.70 | -46.32% |

*Cost note: transaction costs are charged on final-account one-way turnover after offsetting sleeve-level trades. SPY is unchanged by the trading-cost grid because the benchmark is buy-and-hold in this implementation.*

The table does not tell a winner-takes-all story. HAA has the highest gross CAGR, and SPY also earns more than ADAA while taking materially more volatility and drawdown. ADAA is not a return-maximizing strategy. Its case rests on the observed risk-return trade-off and on whether that trade-off survives implementation and robustness checks, not on the highest terminal wealth.

Trading costs materially reduce the headline performance. We use a cost grid set in advance at 0, 5, 10, 25, and 50 basis points per unit of one-way turnover. At 25 basis points, ADAA using the later practitioner weights falls to about 9.36% CAGR and 0.90 Sharpe. The SPY/IEF 60/40 benchmark is about 8.38% CAGR and 0.75 Sharpe at the same cost assumption. Implementation is part of the result, not a footnote to it.

***Table 2B. Uncertainty checks accounting for serial dependence***

| Comparison | Basis | Annualized mean-return difference | HAC(12) p-value | Bootstrap P(mean-return advantage) | Bootstrap P(Sharpe advantage) | Bootstrap P(MDD advantage) |
|---|---|---|---|---|---|---|
| ADAA vs 60/40 | Gross | +2.10 pp | 0.205 | 91.0% | 95.2% | 97.3% |
| ADAA vs 60/40 | 25 bps | +0.83 pp | 0.623 | 67.5% | 78.2% | 96.0% |
| ADAA vs HAA hindsight winner | Gross | -2.12 pp | 0.057 | 2.3% | 38.5% | 86.5% |
| ADAA vs HAA hindsight winner | 25 bps | -1.85 pp | 0.111 | 5.0% | 41.4% | 84.3% |

Note: MDD = maximum drawdown. Bootstrap results use 12-month moving blocks; the reported advantage percentages are resampling frequencies, not p-values or posterior probabilities. An MDD advantage means a shallower maximum drawdown for ADAA.

The block-bootstrap and HAC tests point in the same direction. A 12-month moving-block bootstrap shows a high resampling frequency of ADAA having a shallower maximum drawdown than the SPY/IEF 60/40 benchmark, and the gross Sharpe difference is positive in most bootstrap draws. But the HAC test for the average gross return difference is not conventionally significant. The evidence is therefore stronger for risk-adjusted behavior and drawdown than for a statistically proven mean-return premium.

ADAA also does not beat the best constituent in hindsight. HAA is the strongest full-sample sleeve, at roughly 13.0% CAGR and 1.07 Sharpe gross. That is the model-selection problem in plain view. The best rule is easy to name after the fact; the investable problem is that hindsight is not available when the capital has to be allocated.

Rolling evidence makes the distinction clearer. On 60-month rolling Sharpe, ADAA using the later practitioner weights is above the median constituent in about 98.7% of windows. It appears in the top two among the five sleeves plus the ensemble in about 69% of windows and is never in the bottom two.

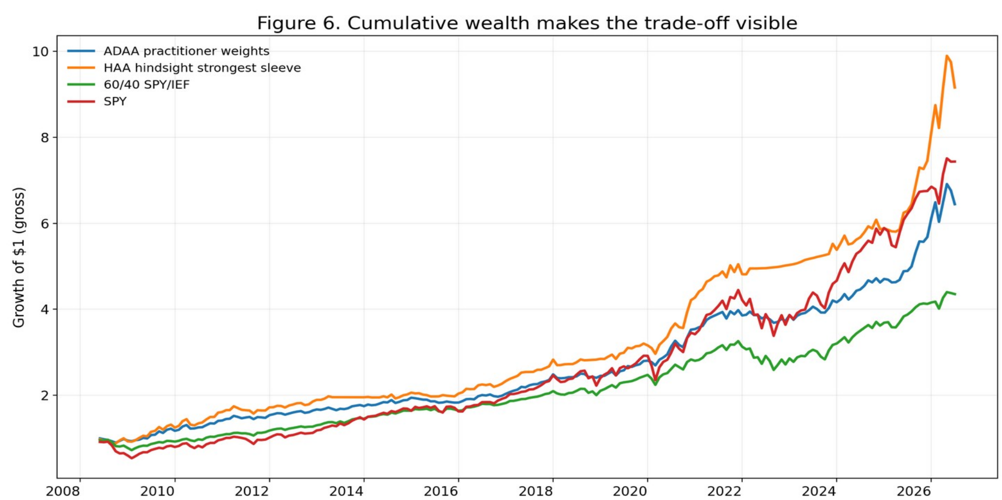


Figure 6. Cumulative wealth makes the trade-off visible. Growth of $1 is shown for gross ADAA using the later practitioner weights, the hindsight-strongest HAA sleeve, the SPY/IEF 60/40 benchmark, and SPY over the common June 2008-July 2026 sample. The chart is descriptive: it shows the path behind the summary statistics, not a claim that ADAA maximizes terminal wealth.

## Drawdown, recovery, and survivable compounding

Drawdown matters because compounding remembers losses. A 20% loss requires a 25% gain to recover; a 50% loss requires a 100% gain. Deep losses therefore create a recovery burden that volatility alone does not show. Research has long treated peak-to-trough loss as a distinct portfolio constraint (Grossman and Zhou 1993; Goldberg and Mahmoud 2017), more recent work emphasizes drawdown duration as a separate dimension of risk (Vedernikov, Liesiö, and Seppälä 2024), and investor-flow evidence suggests that investors respond materially to maximum drawdowns (Riley and Yan 2022).

The goal is not minimum drawdown at any cost. Too much protection can starve participation in growth, and a shallow-drawdown strategy can still compound poorly. The practical objective is survivable drawdown: losses shallow and recoverable enough that the investor can stay invested and the capital base can keep working.

For consistency, drawdown is measured from the running wealth peak with the initial invested wealth of 1.0 included as a valid starting peak. That convention matters in a sample that begins during the 2008 crisis because an immediate loss should count as a drawdown from the capital initially invested.

Under the standard definition, gross ADAA using the later practitioner weights has a maximum drawdown of 10.34%. Recovering from that trough requires an 11.53% gain. The SPY/IEF 60/40 benchmark has a 27.55% maximum drawdown, requiring 38.03% to recover, while SPY falls 46.32% and requires 86.30%. The longest observed underwater spell is 18 months for ADAA, 27 months for 60/40, and 32 months for SPY. These are descriptive sample outcomes, not forecasts of future protection.

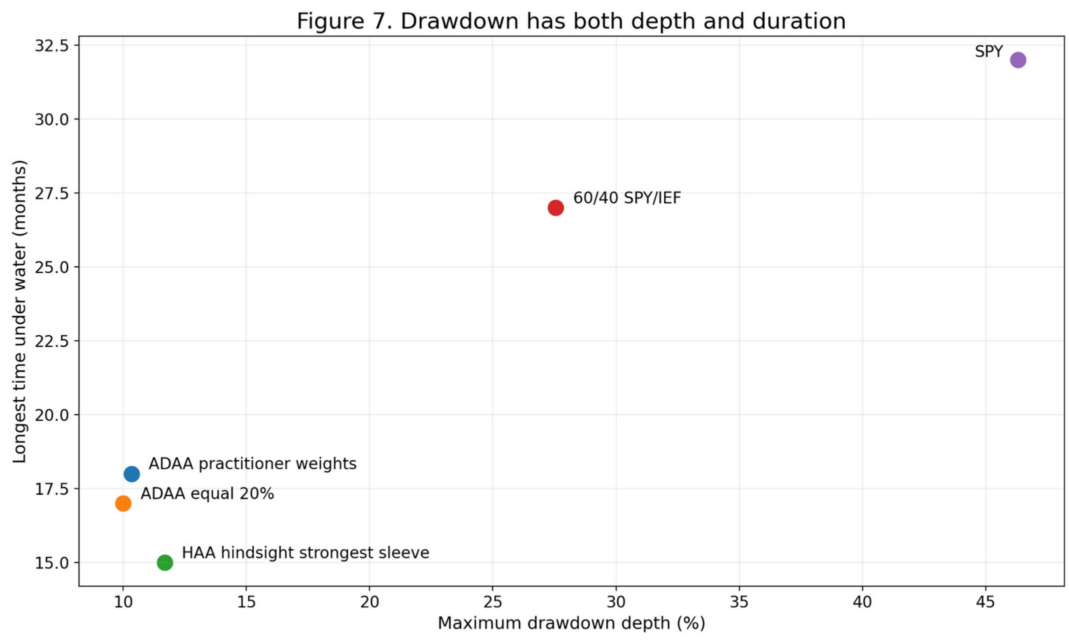


**Figure 7. Drawdown has both depth and duration.** The horizontal axis shows the longest consecutive spell below a prior wealth peak and the vertical axis shows maximum drawdown. The figure uses the standard definition in which initial invested wealth, $W_0 = 1$, is a valid running peak. A portfolio that falls less deeply can still remain underwater for a meaningful period; neither statistic should be interpreted in isolation.

***Table 3. Drawdown anatomy and recovery burden — gross returns***

| Portfolio | Maximum drawdown | Gain required to recover | Longest underwater run | MDD trough | MDD recovery |
|---|---|---|---|---|---|
| ADAA — practitioner weights | **-10.34%** | **+11.53%** | **18 months** | 2008-10 | 2009-05 |
| ADAA — equal 20% | -10.00% | +11.11% | 17 months | 2008-10 | 2009-05 |
| HAA — hindsight strongest sleeve | -11.68% | +13.23% | 15 months | 2008-10 | 2008-12 |
| 60/40 SPY/IEF | -27.55% | +38.03% | 27 months | 2009-02 | 2010-09 |
| SPY | -46.32% | +86.30% | 32 months | 2009-02 | 2011-02 |

The comparison prevents an overly simple story. HAA has the higher CAGR and Sharpe and a shorter longest underwater spell than ADAA, even though its corrected maximum drawdown is slightly deeper. Drawdown control is therefore one part of a survivable architecture, not a standalone objective function. Protection has value only if it leaves enough participation for compounding to continue, so drawdown is treated alongside return, volatility, and turnover rather than minimized in isolation.

### The role of the slow sleeve

LAA makes another point: mechanism should be separated from headline performance.

Removing LAA and rescaling the other four sleeves slightly increases gross CAGR in the equal-weight mechanism test. It also raises volatility and turnover. At a 25-basis-point trading-cost assumption, the version with LAA has the higher CAGR and Sharpe in this sample.

But LAA does not universally improve drawdown. Replacing LAA with RAA, another slow-moving rule, produces an even shallower observed drawdown but slightly lower return. The lesson is not that LAA is the best defensive strategy. It is that the choice of slow anchor changes the ensemble's overall risk-return profile, and persistence itself is a distinct design dimension.

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## 9. Where It Fails

The evaluation is incomplete without explicit failure modes.

### 9.1 Rapid reversal and re-entry lag

The clearest recurring failure pattern is a sharp rebound after a weak prior market.

We mechanically identify the ten worst one-month ADAA returns relative to the SPY/IEF 60/40 benchmark. Seven occur when SPY had fallen over the prior three months and then rose more than 3% in the current month. Examples include March and April 2009, October 2015, January 2019, July and October 2022, and November 2023.

The intuition is straightforward:

> ***A momentum rule can avoid part of a fall and still miss part of the first rebound.***

That lag is the price of waiting for evidence before taking risk again.

### 9.2 Not every stress episode favors ADAA

During the portion of the Global Financial Crisis covered by the sample, ADAA falls about **4.5%**, compared with roughly **23.0%** for the SPY/IEF 60/40 benchmark and **41.9%** for SPY. ADAA also loses less than those benchmarks in the COVID crash and in the 2022 stock-bond stress window.

But 2011 is an important counterexample. ADAA loses about 2.0% over the May-October window defined in advance, while SPY/IEF 60/40 is slightly positive. Adaptation is not a guarantee of winning every risk-off period.

Figure 8 summarizes the asymmetry. The bars show cumulative ADAA return minus 60/40 return for the stress windows defined in advance. The final bar is deliberately different: it averages the seven data-defined worst relative months that also meet the rapid-reversal condition. It is a failure check, not a stress window selected in advance.

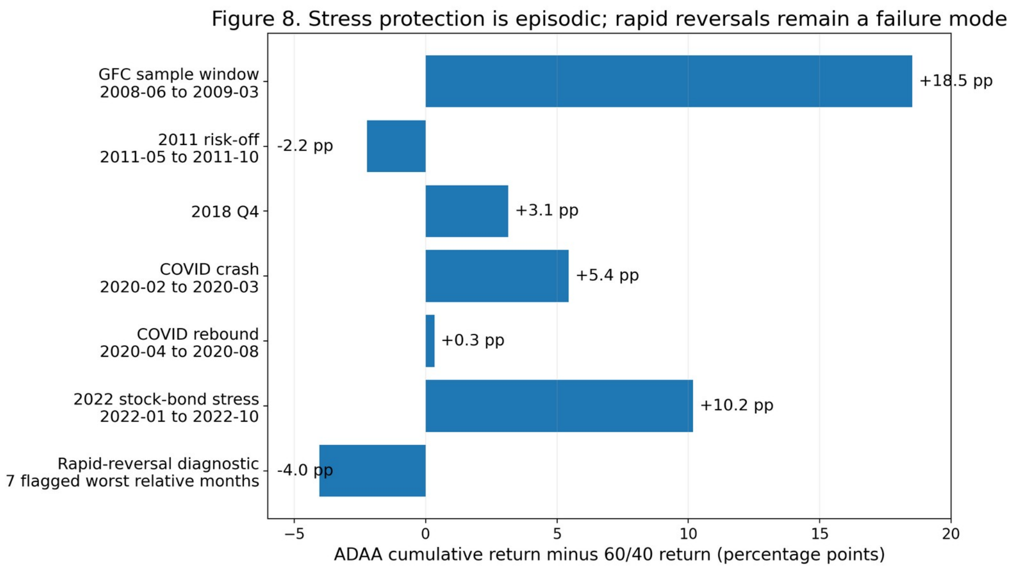


Figure 8. Stress protection is episodic; rapid reversals remain a failure mode. ADAA is substantially ahead of 60/40 in the GFC portion covered by the sample, COVID crash, 2018 Q4, and 2022 stock-bond stress, but not in 2011. The rapid-reversal check averages seven of the ten worst one-month relative outcomes and is negative by about four percentage points.

### 9.3 Slow does not mean safe

LAA loses about **19.6%** in the 2022 stock-bond stress window, even though HAA is close to flat and BAA is positive over the same episode.

Slow is a clock, not a guarantee. Sometimes persistence helps; sometimes it means holding the wrong thing for longer.

### 9.4 Decision diversity does not eliminate common market risk

No stress episode defined in advance has all five sleeves changing target weights simultaneously. That is evidence of decision disagreement, but not immunity from loss. Different decisions can still produce losses together when the underlying asset opportunity set is broadly unfavorable.

Decision Diversification reduces concentration in the decision process; it does not remove common market risk.

## 10. Rules for a Portfolio That Must Evolve

The current sleeves may change. The questions below should still apply. That is the point of treating ADAA as an architecture rather than a fixed recipe.

1. Start with a reason for every rule. A strategy should have an intelligible economic or behavioral rationale. A good backtest is not a rationale by itself.
2. Do not count strategy names. Measure decisions. Ask what the rules own, when they change, and how much risk they take.
3. Use return correlation, but do not stop there. Add holdings overlap, target-weight distance, transition overlap, defensive exposure, concentration, and holding persistence.
4. Look for different clocks. If every rule exits and re-enters at the same speed, several strategies may still amount to one timing bet.
5. Treat ETF substitutions as model changes until proven otherwise. Ask whether the decision role survives the new investable expression.
6. Do not ask one slow sleeve to be a universal hedge. Persistence is a diversification property, not a promise of drawdown protection.
7. Be suspicious of precise optimal weights. Test the neighborhood. A broad plateau is more reassuring than a sharp peak.
8. Charge implementation costs at the final account. Offset underlying buys and sells across sleeves before deciding what the portfolio actually traded.
9. Search for failure mechanisms, not just bad dates. Knowing why the system lost - for example, a rapid rebound after a defensive shift - is more useful than a list of losing months.
10. Make the components replaceable. Treat sleeves as implementations inside a more durable process. Retain, revise, or replace a sleeve through documented tests of role, redundancy, robustness, and implementation. The replacement process should be specified tightly enough to resist rewriting the rule after the outcome is known.

## 11. Korea-Based Investor Extension: Currency Exposure

Because ADAA was developed from the perspective of a KRW-based investor, the USD/KRW exposure problem is practically important even though it sits outside the core Decision Diversification claim.

The historical implementation uses a long-horizon USD/KRW z-score to vary the unhedged USD share among 90%, 50%, and 10%. The historical thresholds and 1,306-trading-day lookback are treated as fixed implementation evidence, not re-optimized after seeing the present sample.

***Table 4. KRW/USD currency-exposure comparison — gross underlying ADAA path***

| Currency exposure | CAGR | Volatility | Maximum drawdown |
|---|---|---|---|
| Hedged proxy | 10.80% | 8.97% | -10.34% |
| Fixed 50% unhedged | 11.99% | 8.41% | -9.39% |
| Fully unhedged | 12.79% | 11.57% | -14.32% |
| **Legacy dynamic 90/50/10** | **13.25%** | **8.17%** | **-6.93%** |

*Note:* These are historical currency-exposure outcomes on the same gross underlying ADAA path, not a forecast or an executable hedge comparison. The full appendix also reports Sharpe, recovery burden, underwater duration, and the 25-basis-point underlying-cost rows.

The comparison is straightforward. The legacy 90/50/10 path has the strongest observed combination of growth, volatility, and drawdown among these four variants, while full USD exposure raises return and risk together. Appendix FX Figure 1 shows the full cumulative wealth paths. The result is historical application evidence, not a claim that the legacy currency rule is ex-ante optimal.

The sensitivity checks matter because the historical thresholds were chosen with past outcomes in view. All sixteen threshold cells defined in advance have a higher observed CAGR than fixed 50% unhedged in this sample, and fifteen have a shallower maximum drawdown.

The historical -0.5/+2.0 pair is not the ex-post CAGR maximum, and the 756-, 1,306-, and 1,827-trading-day windows give similar results on their common evaluation period. That makes the result look less dependent on one narrowly tuned setting, but it does not turn a historically chosen rule into a claim about future performance. Appendix B reports the full check.

### Implementation boundary

The 'hedged' leg is an exposure proxy, not a traded hedge. It omits USD/KRW forward points, NDF bid-ask spreads, collateral, funding, hedge-rebalancing costs, and tax/account effects. The appendix therefore treats this as a currency-exposure extension rather than an executable hedged-return backtest. Drawdown uses the same $W_0 = 1$ convention as the core paper; that correction changes the older hedged-proxy maximum drawdown from 10.16% to 10.34% without changing the monthly return path.

The extension is therefore useful as an applied illustration, while its implementation and inference limits remain separate from the core Decision Diversification claim.

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## 12. Limitations and Boundaries

### Sample and timing

The paper uses ETF-only investable data rather than extending history with weaker proxy series. That keeps the simulation closer to an investable implementation but leaves about eighteen years of common evidence. The June 2008 start is a common-data simulation start, not the beginning of a live ADAA record, and several source rules or variants were published later. Accordingly, all 2008-2026 headline results are historical simulations of the currently specified architecture, not post-publication out-of-sample performance.

### Strategy-family challenge

The historical families were not selected from an exhaustive 2023 census. Appendix A reconstructs a 16-rule reference pool tied to the source landscape documented around that project. Within that pool, the historical five-family set ranks 735 of 4,368 combinations, about the 83rd percentile; a different combination has the highest Decision-Space diversity score, and BAA Aggressive plus BAA Balanced is the closest pair. The four late-added rules test a ranking rule that had already been fixed. This is not historical preregistration, out-of-sample validation in market time, or proof that the Decision-Space measure is uniquely correct. A separate implementation reproduced the late-added rules' target weights to numerical precision without using performance data.

### Sleeve engineering

The current investable sleeves are not untouched copies of their public parents. Parent-versus-variant checks make those changes visible, but they cannot eliminate data-mining risk or the risk that results depend on the exact rule specification. They also reveal a practical tension: improving implementation or diversification inside a sleeve can make the sleeves more similar to one another. The current set should therefore be read as one empirical implementation of an architecture that can evolve, not as permanent components whose future performance is assumed.

### Data, costs, and FX

LAA's unemployment input is aligned to the historically available information state, including the missing October 2025 observation, but the panel uses the present vintage rather than a full ALFRED-style first-release history. Trading costs are tested over a one-way sensitivity grid, while taxes, market impact, fund closure risk, and investor-specific constraints remain outside the model. The KRW/USD appendix is a currency-exposure study, not an executable hedge backtest, because forward/NDF carry, spreads, funding, collateral, hedge-rebalancing costs, and tax effects are not modeled.

### Claim boundary

Decision Diversification is a practitioner label, not a claim that the underlying idea is entirely new. Forecast diversity, model averaging, signal diversification, strategy ensembles, model-speed diversification, and position-overlap analysis are relevant precedents. The contribution here is a DAA-specific operating discipline - What / When / How Much, transition timing, persistence, checks that each sleeve still serves its intended role, weight robustness, and explicit governance of replaceable sleeves - together with a reproducible evidence package.

## 13. Conclusion: What Should Stay When the Strategies Change?

Momentum is useful not because it reveals the next regime, but because it offers a disciplined way to adapt when market evidence changes. Yet one successful idea can be expressed through many reasonable lookbacks, universes, defensive gates, and portfolio rules. A portfolio can therefore contain several strategies and still depend on one broad decision process.

This leads to the paper's central proposal: diversification should extend to the decision process, not stop at the asset list. What a rule owns, when it changes, and how much risk it takes can reveal differences that return correlation misses. The source-to-sleeve audit adds an important complication: improving diversification inside a sleeve can make sleeves behave more alike. Sleeve engineering and portfolio-level diversification therefore have to be governed together.

The empirical record does not support a winner-takes-all conclusion. The later practitioner weights are not treated as weights chosen in advance for this study or as optimal. In the present sample, however, they sit in a broad high-performing region while estimated optimal weights move substantially across rolling windows and bootstrap samples. ADAA does not beat the hindsight-best sleeve, and it fails in identifiable environments such as rapid reversals after defensive shifts. Its case is not guaranteed outperformance, a permanently superior sleeve set, or a timeless rulebook.

The more durable conclusion concerns governance. A multi-rule architecture should diversify decision processes, favor robust neighborhoods over precise historical peaks, and make component changes through documented tests rather than chasing past performance after the fact.

A robust allocation architecture should require fewer things to be exactly right at once. The next regime, the future winning rule, the precise weight vector, and the timing of the next transition are all uncertain.

Drawdown management matters because deep or prolonged losses can interrupt compounding, but minimum drawdown is not the objective. The objective is a transparent, implementable process that can absorb uncertainty, challenge its own components, and evolve without claiming to eliminate uncertainty.

## Appendix A. From a Strategy Zoo to a Decision Portfolio - 16-rule reference-pool check

A fair criticism of ADAA is that several sleeves descend from familiar public allocation strategies. The right response is not to hide those source histories. It is to ask whether the portfolio contains genuinely different decisions - and to allow the test to disagree with the historical design.

We reconstruct a 16-rule common-data 2023 reference pool: AAA, ADM, BAA Aggressive, BAA Balanced, DAA, DM, EAA, FAA, GPM, GTAA, KDA, LAA, PAA, QSF, RAA, and VAA. For rules whose 2023 implementation followed the DB Financial Investment asset-allocation report, we reconstruct the rule logic from the operational recipes cited by the June 2023 project (Kang 2022).

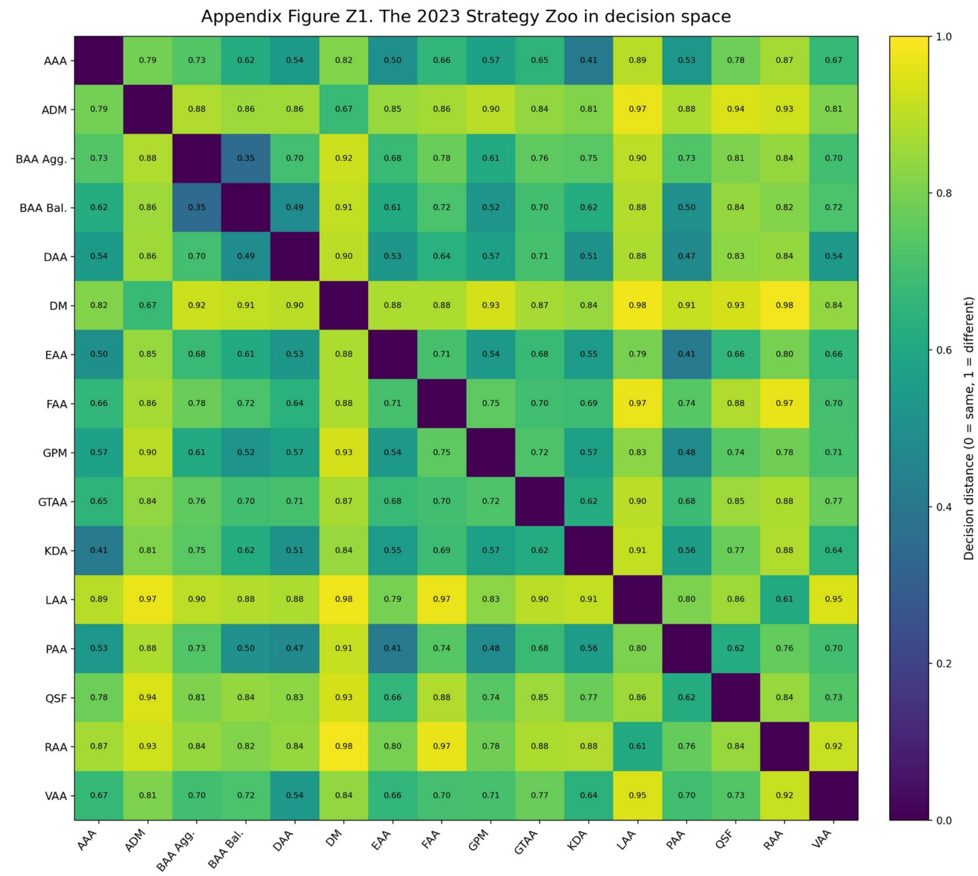
For each pair of rules, Decision-Space compares three parts of the portfolio decision. What: half the average sum of absolute target-weight differences after equivalent ETF exposures are mapped to common economic buckets. Holdings: one minus the average Jaccard overlap of holdings - the share of the combined holdings that the two rules have in common. When: one minus the Jaccard overlap of target-change dates - the share of combined change dates that occur in both rules. Each component runs from 0 (the same) to 1 (maximally different). The pairwise Decision-Space distance is the simple average of What, Holdings, and When. A five-rule set contains ten pairs, so its overall diversity score is the average of those ten pairwise distances. Returns, Sharpe ratios, drawdowns, and other performance statistics are not used. A higher score therefore means more disagreement in portfolio decisions under this specific measure; it does not mean higher expected returns or a statistically better portfolio. All Decision-Space scores in this appendix are computed over the common 216-month target-weight history from July 2008 through June 2026. "Historical 2023 set" refers to the documented five-family composition in the 2023 project, not to a score estimated using only information available at that date.

The Decision-Space definition was fixed before DM, AAA, PAA, and KDA were added to the 16-rule pool. Those four rules therefore provide a late-added check of an already specified ranking rule within this historical study.

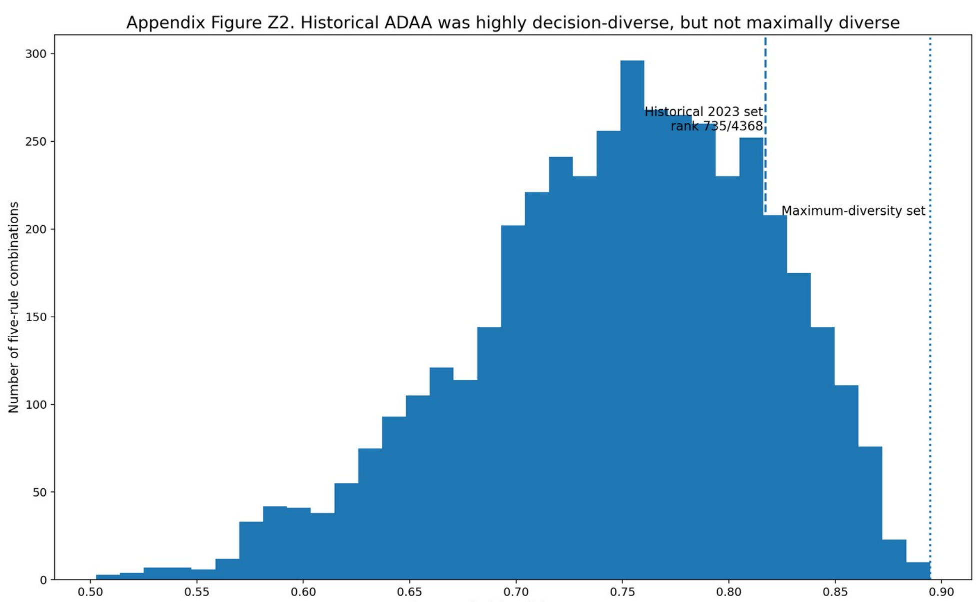
This is not historical preregistration, independent validation of the metric itself, or an out-of-sample test in market time: the market history and the earlier ADAA design were already known.

Among all 4,368 possible five-rule combinations, ADM + DM + FAA + LAA + QSF has the highest Decision-Space diversity score, about 0.895. The historical 2023 family set scores about 0.817 and ranks 735th, around the 83rd percentile. Under this measure, the historical set is more decision-diverse than most feasible sets, but it is not the maximum-diversity set and was not selected by this algorithm.

The most useful challenge result is that BAA Aggressive and BAA Balanced are the closest pair in the 16-rule pool. The performance-free measure therefore identifies redundancy inside the original design rather than simply endorsing it.




**Appendix Figure Z1. The 2023 Strategy Zoo in Decision Space.** Lower values mean more similar portfolio decisions; higher values mean greater disagreement in target allocations, holdings, and transition dates. No strategy-performance statistic enters the matrix.




Appendix Figure Z2. Historical ADAA was highly decision-diverse, but not maximally diverse. Bars count the 4,368 feasible five-rule combinations by Decision-Space score. Higher scores mean greater decision disagreement, not better returns; the figure is not a performance test.

The result is reasonably stable across alternative weightings of the three Decision-Space components that were set in advance. The historical set ranks between 622 and 1,040, about the 76th to 86th percentile. It stays above the median in every specified variant, but none uniquely selects historical ADAA. This supports the broad finding without showing that one particular distance formula is uniquely correct.

HAA remains outside this historical check because it was not part of the candidate set documented in the June 2023 report, not because it was unavailable at the time: Keller and Keuning posted HAA publicly in February 2023.

A separate exploratory check adding HAA places the current source-family set around the 96th percentile of a 17-rule pool. That result is not used to invent an undocumented historical reason for replacing BAA Balanced with HAA.

A second implementation independently reproduced the monthly target weights for DM, AAA, PAA, and KDA to machine precision (maximum absolute difference $< 3 \times 10^{-15}$). It used no performance data and did not alter the Decision-Space ranking rule. Implementation details, including the cross-language check, are documented in the public replication package.

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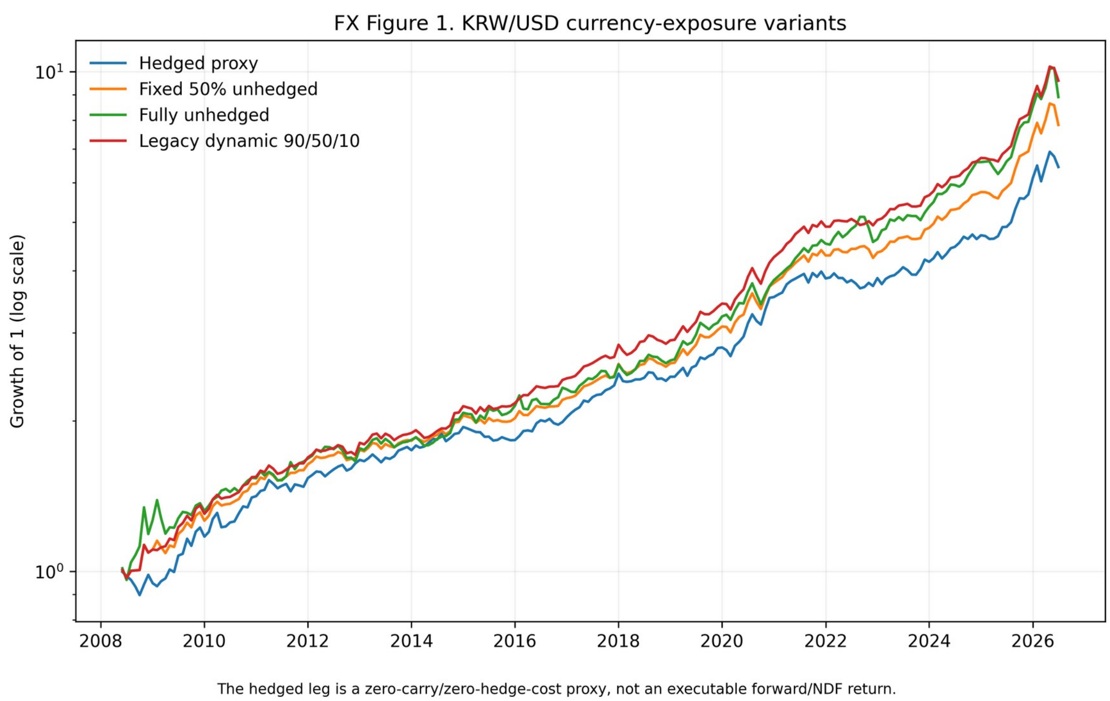
## Appendix B. KRW/USD currency-exposure extension

This appendix isolates the home-currency application from the core global Decision Diversification claim. The historical overlay uses the previously implemented 1,306-trading-day USD/KRW z-score and maps it to three unhedged USD states: 90%, 50%, and 10%. The historical thresholds are kept as historical settings rather than reselected from the present results.

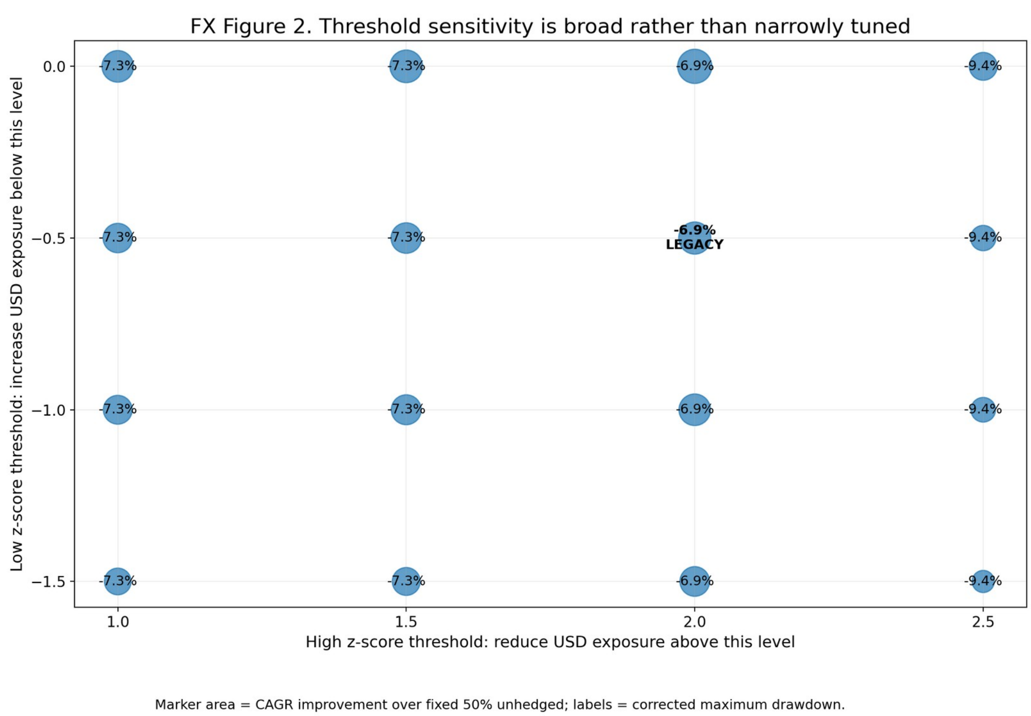



**Appendix FX Figure 1. KRW/USD currency-exposure variants.** The figure compares growth of one under the gross underlying ADAA path with four currency-exposure choices. The “hedged” path is a zero-carry and zero-hedge-cost proxy. It is not an executable forward or NDF return.

***Appendix Table FX-1. KRW/USD currency-exposure performance diagnostics***

| Basis | Currency exposure | CAGR | Volatility | Zero-rate Sharpe | Max drawdown | Gain required to recover | Longest underwater run |
|---|---|---|---|---|---|---|---|
| Gross underlying | Hedged proxy | 10.80% | 8.97% | 1.19 | -10.34% | 11.53% | 18 |
| Gross underlying | Fixed 50% unhedged | 11.99% | 8.41% | 1.39 | -9.39% | 10.36% | 12 |
| Gross underlying | Fully unhedged | 12.79% | 11.57% | 1.10 | -14.32% | 16.71% | 12 |
| Gross underlying | Legacy dynamic 90/50/10 | 13.25% | 8.17% | 1.57 | -6.93% | 7.45% | 7 |
| 25 bp underlying cost | Hedged proxy | 9.36% | 8.97% | 1.05 | -10.75% | 12.05% | 23 |
| 25 bp underlying cost | Fixed 50% unhedged | 10.53% | 8.40% | 1.24 | -9.46% | 10.45% | 16 |
| 25 bp underlying cost | Fully unhedged | 11.32% | 11.56% | 0.99 | -14.49% | 16.95% | 16 |
| 25 bp underlying cost | Legacy dynamic 90/50/10 | 11.78% | 8.16% | 1.41 | -7.03% | 7.56% | 12 |

Note: The 25-basis-point rows apply the underlying final-account transaction-cost path set in advance; no additional currency-hedging implementation cost is modeled. Zero-rate Sharpe is reported only as a descriptive comparison and is not proposed as a Korean-investor benchmark; it should not be compared directly with the core paper's BIL-excess Sharpe.




Appendix FX Figure 2. Threshold sensitivity is broad rather than narrowly tuned. Marker size shows the observed CAGR improvement over fixed 50% unhedged, while labels show corrected maximum drawdown. The legacy threshold pair is marked but not promoted as optimal.

Across the 16-cell grid defined in advance, observed gross CAGR ranges from 12.22% to 13.51% and maximum drawdown from 9.39% to 6.93%.

The long-window sensitivity produces a similarly narrow range on the common post-warm-up sample. Gross CAGR is approximately 12.45% to 12.59% across 756, 1,306, and 1,827 trading days, and each has a 6.93% maximum drawdown.

In a paired 12-month block bootstrap that preserves serial dependence within 12-month blocks, the legacy dynamic rule has a positive CAGR difference versus fixed 50% in 99.76% of draws. Its maximum-drawdown advantage is positive in about 83.2% of draws. These are resampling checks for this historical application; they do not show that the currency rule will dominate in the future.

Two guardrails matter. First, the historical thresholds were chosen with past outcomes in view, so robustness around the rule is more informative than the location of the best cell. Second, an implementable Korean hedging study would require forward/NDF carry, spreads, funding/collateral, hedge-rebalancing costs, taxes, and account-specific constraints. Those ingredients remain outside the present paper.

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## Data and Code Availability

A rights-safe public replication package accompanies this working paper. It contains release-version research code, public source metadata and contracts, derived exhibit data, active figures, and validation summaries. Downloaded raw Yahoo/FRED data, private university materials, proprietary research, and third-party article files are excluded; public-input retrieval steps are documented in the package.

Research dashboard: https://slackquant80.github.io/adaa-slackquant/

Public replication repository: https://github.com/slackquant80/adaa-decision-diversification

Archived replication release (v1.0.2): https://doi.org/10.5281/zenodo.21853533

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